

Name: _____

Solution

Date: _____


SAFETY BAY
 SENIOR HIGH SCHOOL

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Year 10 Mathematics**Investigation 1, 2015****Topic – Exact Values****Validation Section****Total Time:****50 minutes****Weighting:****10% of the year****Equipment:****Exact Values Take-Home Section****No Calculators are to be used**

$\frac{\quad}{71}$ $= \frac{\quad}{\quad} \%$

Question 1 (6 marks: 1, 1, 1, 1, 1, 1)With the following numbers, classify them as either **exact** or **approximate**.

	Number	Exact/Approximate		Number	Exact/Approximate
i	0.1234	E ✓	ii	$2.4\bar{5}$	A ✓
iii	$\frac{2}{11}$	E ✓	iv	π	E ✓
v	$\sqrt{23}$	E ✓	vi	$121.\bar{2}$	A ✓

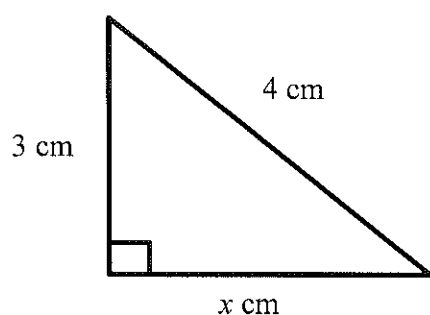
Question 2 (10 marks: 1, 1, 1, 1, 1, 1, 1, 1, 1, 1)Calculate the following. Leave your solution as an **exact** value in its simplest form.

i	$12 \div 3$	4 ✓	ii	$5 \div 10$	$\frac{1}{2}$ ✓
iii	$11 \div 66$	$\frac{1}{6}$ ✓	iv	$\frac{7+2}{21-1}$	$\frac{9}{20}$ ✓
v	$4 \times 5 \div 23$	$\frac{20}{23}$ ✓	vi	$7 \div (6 \times 5)$	$\frac{7}{30}$ ✓

vii	$9 \div 19 \times 2$ $\frac{9}{38} \checkmark$	viii	$\frac{2 \times 7}{26-7}$ $\frac{14}{19} \checkmark$
ix	$\sqrt{144}$ $12 \checkmark$	x	$\sqrt{38+11}$ $7 \checkmark$

Question 3 (3 marks)

Find the **exact** value of x



$$\begin{aligned}
 x^2 &= 4^2 - 3^2 \\
 &= 16 - 9 \quad \checkmark \\
 &= 5 \\
 x &= \sqrt{5} \quad \checkmark \checkmark
 \end{aligned}$$

Question 4 (16 marks: 2, 2, 2, 2, 4, 4)

Simplify the following

a. $3\sqrt{3} + 4\sqrt{3}$

$$7\sqrt{3} \quad \checkmark \checkmark$$

b. $11\sqrt{7} - 7\sqrt{7}$

$$4\sqrt{7} \quad \checkmark \checkmark$$

c. $12\sqrt{19} + \sqrt{19} + 5\sqrt{19}$

$\checkmark\checkmark 18\sqrt{19}$

d. $2\sqrt{5} - 9\sqrt{5} + 3\sqrt{5}$

$\checkmark\checkmark -4\sqrt{5}$

e. $2\sqrt{17} + 13\sqrt{3} + 11\sqrt{17} + 5\sqrt{3}$

$\checkmark\checkmark 13\sqrt{17} + 18\sqrt{3}$

f. $5\sqrt{7} - 7\sqrt{21} + 6\sqrt{7} - 5\sqrt{21}$

$\checkmark\checkmark 11\sqrt{7} - 12\sqrt{21}$

Exercise 7 (12 marks: 2, 2, 2, 2, 2, 2)			
Factorise the following so that one factor is a square number			
a.	20	b.	27
	4×5 $\checkmark\checkmark$		9×3 $\checkmark\checkmark$
c.	125	d.	28
	25×5 $\checkmark\checkmark$		4×7 $\checkmark\checkmark$
e.	36	f.	147
	9×4 $\checkmark\checkmark$		49×3 $\checkmark\checkmark$

Exercise 8 (18 marks: 3, 3, 3, 3, 3, 3)

Simplify the following square roots

a.	$\sqrt{24}$ $= \sqrt{4 \times 6}$ $= \sqrt{4} \times \sqrt{6} \quad \checkmark$ $= 2\sqrt{6} \quad \checkmark\checkmark$	b.	$\sqrt{18}$ $= \sqrt{9 \times 2}$ $= \sqrt{9} \times \sqrt{2} \quad \checkmark$ $= 3\sqrt{2} \quad \checkmark\checkmark$
c.	$\sqrt{45}$ $= \sqrt{9 \times 5}$ $= \sqrt{9} \times \sqrt{5} \quad \checkmark$ $= 3\sqrt{5} \quad \checkmark\checkmark$	d.	$\sqrt{300}$ $= \sqrt{100 \times 3}$ $= \sqrt{100} \times \sqrt{3} \quad \checkmark$ $= 10\sqrt{3} \quad \checkmark\checkmark$
e.	$\sqrt{63}$ $= \sqrt{9 \times 7}$ $= \sqrt{9} \times \sqrt{7} \quad \checkmark$ $= 3\sqrt{7} \quad \checkmark\checkmark$	f.	$\sqrt{162}$ $= \sqrt{81 \times 2}$ $= \sqrt{81} \times \sqrt{2} \quad \checkmark$ $= 9\sqrt{2} \quad \checkmark\checkmark$